

INTRODUCTION TO DIGITAL FORENSICS

- **Digital Forensics** is the branch of forensic science focused on the **identification, preservation, analysis, and presentation** of data stored or transmitted in digital form so it can be used as evidence in legal or organizational proceedings.
- It is commonly applied in:
 - Criminal investigations (e.g., hacking, fraud)
 - Corporate investigations (e.g., insider threats)
 - Incident response (e.g., ransomware attacks)
 - Civil litigation (e.g., intellectual property theft)

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- **Core Objectives**

- **Identify** digital evidence
- **Preserve** evidence without altering it
- **Analyze** data to reconstruct events
- **Document & Present** findings in court or reports

- **Key Characteristics of Digital Evidence**

- Fragile (can be altered easily)
- Volatile (some data disappears when power is lost)
- Requires specialized tools
- Must maintain chain of custody

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- **Simple Example**

A company suspects an employee of stealing confidential files:

- Investigators create a **forensic image** of the employee's laptop.
- Recover deleted files.
- Analyze USB device history.
- Examine email logs.
- Produce a report showing the files were copied to an external drive.

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Computer Forensics: Focuses on evidence from computers, laptops, and storage devices.

- **Areas Covered**

- Hard drives
- SSDs
- File systems
- Deleted file recovery
- User activity logs
- Application artifacts

- **Common Artifacts**

- Browser history
- Recycle Bin data
- Registry entries (Windows)
- Log files
- Metadata (file timestamps)

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- **Example**
- In a fraud case:
 - A suspect deletes financial spreadsheets.
 - Investigators recover deleted files.
 - Analyze file metadata.
 - Show that spreadsheets were altered before deletion.

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- **Network Forensics:** Focuses on monitoring and analyzing network traffic.
- **Goals**
 - Detect intrusions
 - Trace attackers
 - Analyze data exfiltration
 - Investigate malware communication
- **Data Sources**
 - Firewall logs
 - Router logs
 - Packet captures (PCAP)
 - IDS/IPS logs
 - DNS logs

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Example

- A company suffers a ransomware attack:
- Analysts examine firewall logs.
- Identify suspicious outbound traffic.
- Trace connection to a malicious IP.
- Determine the initial infection vector (phishing email).

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- **Memory Forensics (Volatile Memory Forensics)**

- Focuses on analyzing RAM contents.

- **Why It's Important**

- RAM contains:
 - Running processes
 - Encryption keys
 - Network connections
 - Unsaved data
 - Malware running only in memory

- **Characteristics**

- Extremely volatile
 - Must be captured before system shutdown

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- **Example**

- A system infected with fileless malware:
- No malicious files found on disk.
- RAM dump reveals suspicious process.
- Extracted encryption keys from memory.
- Identified command-and-control server.

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- **Cloud Forensics**

- Focuses on investigations involving cloud environments.

- **Challenges**

- Multi-tenant environments
- Limited physical access
- Jurisdiction issues
- Log availability

- **Evidence Sources**

- Cloud logs
- Access logs
- Virtual machine snapshots
- Cloud storage metadata

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What is Cybercrime?

- Cybercrime refers to criminal activities involving computers, networks, or digital devices.

Categories of Cybercrime

1. Crimes Against Individuals

- Identity theft
- Online harassment
- Phishing scams

2. Crimes Against Organizations

- Ransomware attacks
- Data breaches
- Insider threats

3. Crimes Against Government

- Cyber espionage
- Cyber terrorism
- Critical infrastructure attacks

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- **Common Cybercrimes**

- **Phishing**

- Fake emails designed to steal credentials.
- Example:
An attacker sends fake banking emails asking users to reset passwords.

- **Ransomware**

- Malware that encrypts files and demands payment.
- Example:
An attacker encrypts hospital systems and demands cryptocurrency payment.

- **Insider Threat**

- An employee steals intellectual property before leaving a company.

- **Distributed Denial of Service (DDoS)**

- Attack that overwhelms a server with traffic.
- Example:
Botnet floods a company website, making it unavailable.

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- **Digital Evidence Lifecycle**

- The **Digital Evidence Lifecycle** refers to the structured process used to handle digital evidence from start to finish.

- **Identification**

- Recognizing potential sources of digital evidence.
- **Examples**
 - Computers
 - Mobile devices
 - Cloud accounts
 - USB drives
 - Network logs

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- **Preservation**

- Ensuring evidence remains unchanged.
- **Key Practices**
- Create forensic image (bit-by-bit copy)
- Use write blockers
- Calculate hash values (MD5/SHA-256)
- Document chain of custody

- **Example**

- Before analyzing a hard drive:
- Investigator creates an image.
- Computes hash value.
- Verifies integrity.

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- **Collection**

- Acquiring digital evidence properly.

- **Methods**

- Disk imaging
- RAM capture
- Log export
- Network traffic capture

- **Examination**

- Processing and extracting relevant data.

- **Tasks**

- Recover deleted files
- Keyword searching
- Timeline analysis
- Decrypting files (if authorized)

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- **Analysis**
 - Interpreting evidence to reconstruct events.
- **Questions Answered**
 - Who performed the action?
 - When did it happen?
 - What data was accessed?
 - How was the attack executed?
- **Documentation**
 - Recording every action taken.
 - Includes:
 - Tools used
 - Hash values
 - Procedures followed
 - Findings

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- **Presentation**

- Delivering findings in:
 - Written forensic report
 - Court testimony
 - Executive summary

- Report must be:

- Clear
- Objective
- Technically accurate
- Legally defensible

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- **Chain of Custody**

- A documented record showing:
- Who collected evidence
- When it was collected
- Who handled it
- Where it was stored
- Maintains integrity and admissibility in court.

Area	Focus	Example
Computer Forensics	Disk & file analysis	Recover deleted files
Network Forensics	Traffic & logs	Trace hacker IP
Memory Forensics	RAM analysis	Extract encryption key
Cloud Forensics	Cloud logs & VMs	Investigate data breach

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- **Quick Review Questions**

- Define digital forensics.
- Explain differences between computer and memory forensics.
- What is volatile data? Give examples.
- List stages of the digital evidence lifecycle.
- Why is hashing important in digital investigations?

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- **Chain of custody (CoC)** is the documented and unbroken transfer of evidence from the time it is collected until it is presented in court. In digital forensics, it ensures that digital evidence (e.g., hard drives, USB devices etc) has not been altered, tampered with, or contaminated.
- **Purpose**
 - Maintain **integrity** of digital evidence
 - Establish **authenticity**
 - Ensure **admissibility in court**
 - Provide **accountability** for everyone who handled the evidence
- Without a proper chain of custody, evidence can be challenged and potentially ruled inadmissible.

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- **Key Principles**

- **Continuity** – Evidence must be accounted for at all times.
- **Documentation** – Every action taken on the evidence must be recorded.
- **Integrity Protection** – Use of hashing and write-blockers to prevent alteration.
- **Secure Storage** – Controlled access and secure evidence lockers.
- **Transparency** – Clear identification of handlers and actions performed.

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- **Steps in Chain of Custody**
- **1. Identification**
- Recognize potential digital evidence (computers, mobile phones, servers, cloud accounts).
- Assign a unique evidence ID.
- **2. Collection**
- Photograph and document the scene.
- Use proper forensic tools.
- Record:
 - Date and time
 - Location
 - Collector's name
 - Description of item

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3. Acquisition

- Create a **forensic image** (bit-by-bit copy).
- Use write-blockers to prevent modification.
- Generate cryptographic hash values (e.g., MD5, SHA-256) before and after imaging to confirm integrity.

4. Preservation

- Store original evidence securely.
- Analyze only the forensic copy, never the original.

5. Documentation of Transfers

- Each transfer must include:
 - Date and time
 - From whom
 - To whom
 - Purpose of transfer
 - Signatures

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- **Presentation in Court**

- Demonstrate documented continuity.
- Show matching hash values to prove integrity.
- Provide testimony explaining handling procedures.

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- **Common Challenges**

- Improper documentation
- Unauthorized access
- Failure to use write-blockers
- Hash mismatches
- Gaps in transfer logs
- Cloud and remote data jurisdiction issues

- **Legal Relevance**

- Courts rely on chain of custody to determine:
 - Whether evidence is authentic
 - Whether it has been tampered with
 - Whether handling procedures were reliable

- Failure in maintaining chain of custody can lead to exclusion of evidence under evidentiary rules.

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- **Volatile data** is information that is **lost when a system is powered off** or restarted.
- **Characteristics:**
 - Temporary
 - Stored in RAM or cache
 - Changes frequently
 - Must be collected quickly
- **Examples:**
 - Running processes
 - Active network connections
 - Logged-in users
 - RAM contents
 - Clipboard data
 - Encryption keys in memory

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- **Non-volatile data** remains stored even after the system is powered off.
- **Characteristics:**
 - Persistent storage
 - Stored on disks or external media
 - Can be recovered later
- **Examples:**
 - Hard drive/SSD data
 - Log files
 - Emails
 - Documents
 - Installed applications
 - Registry files

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First Responder Priorities in Digital Forensics

- A **first responder** is the first person to arrive at a digital crime scene.

1. Ensure Safety

- Ensure physical safety of people.
- Secure the scene.

2. Follow Order of Volatility

- Collect evidence from **most volatile to least volatile**:
- CPU registers / cache
- RAM
- Network connections
- Running processes
- Disk data
- Backups

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3. Preserve Evidence

- Do not alter data unnecessarily.
- Avoid shutting down system immediately (unless required).
- Use write blockers when imaging drives.

4. Document Everything

- Photograph the scene.
- Record date, time, and actions taken.
- Maintain chain of custody.

5. Isolate the System

- Disconnect from network (to prevent remote tampering).
- Do not power off unless necessary.

6. Acquire Evidence Properly

- Capture memory if system is live.
- Create forensic image of storage media.
- Verify integrity using hash values (MD5/SHA).

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